Mechanistic Dissection of PI-GNN for Graph Coloring

Final experiment plan (merged version, second revision after review)

0. Scope decisions and what changed

This version keeps the compact structure of the updated plan and restores a few specifics from the earlier detailed version.

- **Scope:** three primary research questions and exactly three core experiments. Warm starts and mean-field annealing are secondary case studies. The Shapley component grid is dropped.
- **Gating:** the heavy mechanism work (fixed preconditioner, update alignment, architecture checks) runs only if PI-GNN differs from free logits. The cheap diagnostics (Dirichlet energy, movement from initialisation) are logged in every Experiment 3 run, so a null result still comes with an explanation.
- **Restored:** the spectral argument for the preconditioner; named prior work in the body, including Vanderbush & Weber's free-logit ablation; an explicit statement that the search protocol is held fixed; a definition of D-CE.
- **Not restored:** the control that gives TabuCol the same warm start and the same time. Add it only if time permits.
- **Added after review:** the soft best-response gap g(v), normalised by degree and paired with normalised row entropy, as a first-order coordinate-stationarity diagnostic (Experiment 1); stratified DIMACS reporting with descriptive p-values (Section 4); two further null controls in Experiment 2 (colour coordinates permuted within each row, and D-CE with a random vertex order); a corrected Dirichlet-energy caveat (4C).
- **Corrected after review:** the earlier claim that Â² is a low-pass filter that damps the components colouring needs is dropped. Diagnostic 4A now uses the actual induced preconditioner and measures where colouring information lies spectrally; the self-loop prediction is restricted to bipartite or strongly alternating structure (4D); 4B is run with plain gradient descent as well as the optimiser actually used.
- **Added in the second review:** free logits start from PI-GNN's exact initial logits, with a random-initialisation arm as a robustness check (Experiment 3); k = χ−1 is the primary setting for Experiments 1 and 3, with k = χ secondary (Section 4); 4A is reworded to separate continuous-time gradient flow from finite-step descent and to flag that M_G equals Â² only when the embedding dimension is at least n; the 4D self-loop prediction is stated for the Â² approximation and then tested against the actual induced preconditioner.

1. Goal and scope

The goal is not a new colouring heuristic and not a claim that PI-GNN beats classical solvers. We ask:

Which parts of PI-GNN are responsible for its behaviour when it optimises the Potts relaxation for graph coloring?

We separate PI-GNN into three normally entangled components: (1) the optimisation and parameterisation that produce a soft colouring P, (2) the information actually contained in P, and (3) the decoding or local search that turns P into a hard colouring. The Potts objective is held fixed in the main study because existing experiments already show that changing Potts to the Erdős objective has little effect under a controlled protocol.

Scope: per-instance PI-GNN only. HWS's inductive results and geocolor's M-IND work appear as pilot evidence and in the discussion, not as new experiments. DSATUR, randomised DSATUR and TabuCol are reference points, not the contribution.

2. Primary research questions

- **RQ1, endpoint dynamics.** Does PI-GNN approach a hard colouring that is single-flip optimal as training proceeds? How do endpoint quality and local optimality change with training budget?
- **RQ2, GNN contribution.** Does the GNN parameterisation reach better minima than directly optimising one logit vector per vertex? If it does, can a fixed graph-dependent preconditioner reproduce the difference?
- **RQ3, decoding and information in P.** How much of the final result comes from graph-specific information in the learned soft output P, and how much is supplied by the discrete decoder or local search applied afterwards?

Secondary analyses serve only to interpret these questions: warm starts as a case study for RQ1 and RQ2, and mean-field annealing as a lineage control for RQ2.

3. Components and what is already known

Component	Variants	Role and prior work	Priority
Parameterisation	PI-GNN; free logits; no-message-passing control; fixed graph-preconditioned descent	Central test of whether the GNN itself changes optimisation. Vanderbush & Weber already ablated depth 0 (free logits) on Erdős–Rényi graphs; GNN-free relaxations exist (Goudet et al., qudit-inspired optimisation, PQQA).	Core
Decoder	Argmax; sampling; D-CE; argmax + vertex descent; null-P controls	Central test of whether performance comes from P or from the discrete procedure. Karalias & Loukas and Wang et al. (2022) show that entry-wise concave relaxations round without losing cost; UCOM2 studies derandomisation only for robust colouring.	Core

Component	Variants	Role and prior work	Priority
Endpoint and local search	Single-flip optimality; strict-improvement vertex descent	Tests whether PI-GNN reaches or merely approaches basic local minima.	Core
Objective	Potts; optional small Erdős confirmation	geocolor TC1–TC3 already show a tie when the rest is controlled. Motivation only.	Supporting
Search protocol	Fixed k throughout	geocolor shows q-descent has a large effect (71/49/0 against a single run). We hold it constant so that parameterisation and decoding are compared cleanly; q-descent is out of scope.	Out of scope
Initialisation	Random; selected DSATUR and TabuCol warm starts	Case study explaining earlier no-op warm-start results; no broad sweep. Wang & Li proposed greedy initialisation; Vanderbush & Weber studied prior strength.	Secondary
Lineage	Mean-field annealing; vertex descent; TabuCol	Historical and simple-optimiser comparison after the core results.	Secondary

What we do not claim:

- That the link between Potts local minima and single-vertex local search is new. It follows from the multilinear structure used in the relaxation and rounding work of Karalias & Loukas and Wang et al.
- That mean-field annealing for colouring is new (Peterson & Söderberg 1989; Di Blas, Jagota & Hughey 2002).
- That PI-GNN gets trapped in local optima, or that classical heuristics beat neural colouring, as new observations (Angelini & Ricci-Tersenghi; Boettcher; Krutský et al. analyse PI-GNN training dynamics for QUBO problems).
- That comparing a GNN with free logits is new in itself (Vanderbush & Weber).

What is potentially new: the controlled mechanistic decomposition on structured colouring benchmarks: endpoint-versus-training-length analysis; decoder null controls that test whether P is informative; and the ladder from direct free-logit optimisation through fixed graph-preconditioned descent to the full PI-GNN.

4. Shared protocol

Instances and baselines

- **Primary test set:** Semyon's 48 DIMACS graphs with χ proven by color6, minus graphs with $\chi \leq 2$, where $k = \chi - 1 = 1$ leaves a single colouring.
- **Primary regime: $k = \chi - 1$.** At $k = \chi$ a successful run has zero conflicts and is trivially single-flip optimal, so it says nothing about local-minimum quality. At $k = \chi - 1$ zero conflicts are impossible, so every method settles in a nontrivial landscape. Experiments 1 and 3 therefore use $k = \chi - 1$ as primary and report $k = \chi$ (success rates, and the unsuccessful runs) as secondary. color6 gives χ but not the minimum conflicts at $\chi - 1$, so measure quality against the best TabuCol result, not a true optimum. Semyon's E3 graphs are a natural pilot, but they were partly selected for hardness, so the main set is all graphs with $\chi \geq 3$.
- Because DIMACS is heterogeneous, report both aggregate and stratified results by graph family, size, density and baseline difficulty, where sample sizes permit. Treat paired p-values on this benchmark as descriptive evidence, not as estimates from an IID population of graphs.
- **Tuning and diagnostic set:** disjoint synthetic graphs of similar size and density (Erdős–Rényi, Watts–Strogatz, random regular, planted k-colourable). Avoid instances where randomised DSATUR already solves nearly everything.
- **Search protocol:** k is fixed in all comparisons, as in PI-GNN. q-descent and restarts are not varied.
- **Baselines:** DSATUR, randomised DSATUR and TabuCol, as reference points for quality and compute.

Fairness rules

- **Tune each method separately** on the same disjoint tuning set. Equal learning rates are not necessarily fair; the goal is equal tuning effort. Record and report the grids. Vanderbush & Weber found the published PI-SAGE learning rates too large to converge.
- **Start from the exact same initial logits.** Whenever PI-GNN is compared with free-logit or preconditioned optimisation, set Z_0 for the free-logit arm (and for rung D) equal to PI-GNN's initial output $f_\theta(G, X)$, taken in eval mode so it is deterministic, for every graph and seed. Matching only the logit scale would leave a confound: PI-GNN's initial logits already carry graph-induced correlations from message passing (smoothed across neighbours, which may be the wrong direction for colouring), so a later difference could come from the starting point rather than the dynamics. Use the same k, decoder, repair rule and stopping criterion. Report free logits from random initialisation (matched scale) as a secondary robustness arm.
- **Report both matched optimisation steps and matched wall-clock time.** Free logits are much cheaper per step, so both views are necessary.
- **Save P at several training budgets** (for example 100, 1,000 and 10,000 epochs, plus the longest feasible run). Endpoint claims must be shown as a function of training length, not from one short budget. Earlier PI-GNN runs here used 100–800 epochs against the papers' 100,000.

- **Seeds:** 5 for the final comparisons. Exploratory runs may use fewer but cannot support headline claims.

Definitions and statistics

- **Single-flip optimality:** no recolouring of one vertex strictly lowers the conflict count. Only vertices incident to a conflict can improve, so only they need checking.
- **Vertex descent:** steepest strict-improvement recolouring on conflicted vertices with random tie-breaking. Plateau (sideways) moves, if tested, are reported separately. TabuCol is essentially this plus plateau moves and a tabu list.
- **D-CE** (derandomised conditional-expectation decoding, from the Erdős line): visit the vertices in saturation order (most distinctly coloured neighbours first, as in DSATUR) and fix each to the colour that minimises the expected number of conflicts, given the colours already fixed and the soft distributions of the remaining vertices. The resulting conflict count is at most the Potts loss of P. Match the order to geocolor's implementation.
- **DSATUR at fixed k:** run DSATUR with at most k colours, giving any vertex with no free colour its least-conflicting colour.
- **Statistical unit:** the graph. Average seeds within each graph, then run paired comparisons across graphs. Report win/tie/loss counts, mean and median differences, confidence intervals, and a sign or Wilcoxon test where appropriate.

5. Core experiments

Experiment 1: endpoint dynamics over training length (RQ1)

- **Primary regime:** $k = \chi - 1$, where zero conflicts are impossible and the question of reaching a one-flip minimum is meaningful. $k = \chi$ is reported secondarily, to compare successful and unsuccessful feasibility runs. Quality is measured against the best TabuCol result.
- For each PI-GNN run and each saved training budget, decode P by argmax, count conflicts, test single-flip optimality, and run strict-improvement vertex descent if needed.
- Record conflicts before and after descent, the number of improving moves, the fraction of conflicts removed, whether the colouring was already single-flip optimal, and the descent runtime.
- Also measure a soft P-space coordinate-wise best-response gap at every saved budget. For each vertex v and colour c, let $q(v,c)$ be the sum over the neighbours u of v of $p(u,c)$, and define $g(v) = \Sigma_c\, p(v,c)\, q(v,c) - \min_c q(v,c)$, which is never negative. Normalise by degree, $\tilde{g}(v) = g(v) / \deg(v)$, excluding isolated vertices, and pair it with the normalised row entropy $\tilde{H}(v) = -\Sigma_c\, p(v,c) \log p(v,c) / \log k$. Report the joint distribution of $(\tilde{g}, \tilde{H})$, not two separate means.
- **How to read the pair:** g is a first-order coordinate-stationarity diagnostic, not an optimality certificate. It is exactly zero at the uniform symmetric point ($\tilde{H} = 1$), which is the known trap. So $\tilde{g} \approx 0$ with low entropy indicates a near-hard, coordinate-wise optimal output, while $\tilde{g} \approx 0$ with $\tilde{H}$

≈ 1 indicates the symmetric saddle. For a hard colouring, g(v) equals the vertex's conflicts minus its best single-flip conflicts, so it agrees with the hard test.

- Compare descent from PI-GNN argmax with descent from a random colouring and from DSATUR at the same fixed k. A weak TabuCol start may be included only if already available.
- **Primary figure:** training budget against (i) the fraction of single-flip-optimal argmax outputs, (ii) conflicts before and after vertex descent, and (iii) the joint $(\tilde{g}, \tilde{H})$ distribution as a companion panel.
- **Caveat:** the Potts lemma concerns local minima in P-space (for a fixed palette the loss is linear in each vertex's distribution, so every strict local minimum over the product of simplices is a hard, single-flip-optimal colouring). PI-GNN optimises parameters θ, so a stationary point in parameter space need not be a local minimum in P-space. The endpoint connection is therefore empirical, not automatic.

Interpretation: if local optimality rises with budget and descent gains vanish, PI-GNN is approaching ordinary discrete local minima. If substantial descent gains remain even after long training, PI-GNN often stops before reaching even a basic one-flip endpoint.

Experiment 2: decoder versus local search, with null controls (RQ3)

- Freeze the same trained P and decode it with: argmax; sampling K colourings; D-CE; argmax plus vertex descent; optionally sampling plus vertex descent. Do not retrain between decoder variants.
- Run the same decoders on null controls: (a) a uniform-plus-noise P; (b) P from an untrained, randomly initialised GNN; (c) the trained P with its rows permuted across vertices (same marginal distribution, no alignment with the graph); (d) the trained P with the colour coordinates independently permuted within each vertex row (each vertex keeps its confidence and entropy, but colour identities are no longer coordinated across vertices); and (e) D-CE with a random vertex order instead of saturation order.

Result	Interpretation
D-CE ≈ argmax + descent	Most of the decoder gain is explained by simple discrete local improvement
D-CE(trained) > argmax + descent, and > D-CE(null or permuted)	The learned soft output carries graph-specific information that argmax loses
D-CE(trained) ≈ D-CE(null or permuted)	The decoder itself behaves like the solver; the learned P contributes little
D-CE only slightly better but much slower	The sophisticated decoder may not be justified computationally

- **Metrics:** conflicts before and after repair, valid-colouring rate, colours used where relevant, decoder runtime, pairwise win/tie/loss.

- **Existing evidence:** geocolor B5 on 132 PI-GNN outputs: mean conflicts 53.4 (argmax) against 5.0 (D-CE), 88 wins, 44 ties, 0 losses; the guarantee conflicts ≤ L_Potts held on all 132. B5 motivates this experiment, but the null controls are the crucial new diagnostic.

Experiment 3: does the GNN parameterisation matter? (RQ2)

Rung	Parameterisation	What it isolates
A	PI-GNN: P = softmax(f_θ(G, X))	The full method
B	Free logits: P = softmax(Z), Z in R^(n × k)	Direct optimisation of the same relaxation, without a GNN
C	No-message-passing control: learned node embeddings followed by an MLP (secondary rung)	Over-parameterisation without graph aggregation. Check whether our PI-GNN's input embeddings are themselves trainable; if so, this rung isolates that.
D	Fixed graph-preconditioned descent: Z(t+1) = Z(t) − η M_G ∇_Z L, with M_G the linearised induced preconditioner of 4A (gated extension)	Whether message passing mainly acts as a fixed graph filter or preconditioner

- **Mandatory comparison:** A versus B, with B started from A's exact initial logits (paired by graph and seed) and k = χ−1 as the primary setting. At k = χ the outcome is mostly binary success (PI-GNN solved only 11 of 48 graphs in Semyon's E2, at small budgets), whereas conflicts at χ−1 are graded and give the paired tests more power. Three arms: PI-GNN (A); free logits from the same Z_0 (B); free logits from random initialisation (B-rand). A against B isolates the dynamics; B against B-rand isolates the initialisation. Tune each method separately and report equal-step and equal-time results.
- **Noise control:** the paper-median PI-GNN configuration uses dropout 0.328, which injects noise into PI-GNN's updates that free logits lack. Report PI-GNN with and without dropout, or add a noise-injection control for free logits, so an advantage is not attributed to message passing when it comes from the noise.
- **Decoder:** use the same decoder across rungs. Report argmax first as the measure of raw optimiser quality; D-CE is then an end-to-end secondary view.
- **Metrics:** Potts loss trajectory, hard conflicts against iteration and time, best and final conflicts, time to best, fraction of single-flip-optimal endpoints, endpoint diversity across seeds.
- **Always logged (ungated):** in every run, record the Dirichlet energy of P and the parameter and logit distance from initialisation (diagnostics 4C and 4E below). They cost almost nothing and give a mechanism-level result even if A ≈ B.

Result	Reading
B ≈ A	The GNN adds little beyond reparameterising the relaxed

Result	Reading
	variables
B > A	Message passing may actively hurt, consistent with the heterophilous nature of colouring
A > B	The GNN changes the optimisation enough to justify the gated mechanistic follow-up
D ≈ A while B differs	Graph-structured coupling, rather than a learned nonlinear representation, explains much of the GNN effect
A > B and D	Full nonlinear, evolving GNN dynamics contribute beyond direct optimisation and a fixed filter
B-rand differs from B	The starting point, not the dynamics, drives part of the difference

6. Mechanistic diagnostics

Always logged during Experiment 3: 4C and 4E. **Gated:** 4A, 4B and 4D run only if Experiment 3 shows that PI-GNN behaves meaningfully differently from free logits. The course project is complete without the gated parts if Experiments 1–3 are clean.

- **4A, induced preconditioner (gated).** Under continuous-time gradient flow, $d\theta/dt = -\nabla_\theta L$, the logits satisfy exactly $dZ/dt = -K(t)\,\nabla_Z L$, where $K(t) = J(t)\,J(t)^T$ is the Gram matrix of the Jacobian of the logits with respect to the parameters. For finite-step gradient descent this gives only the first-order approximation $\Delta Z \approx -\eta\,K(t)\,\nabla_Z L$, because the network is nonlinear in θ. Adam gives a different first-order update, approximately $\Delta Z \approx -\eta\,J\,D_t\,J^T\,\nabla_Z L$ with D_t a diagonal per-parameter rescaling. Treating K as constant additionally requires lazy training, which 4E checks. In the simplest linearised case (one linear layer, fixed features X), K acts on nodes as $M_G = \hat{A}\,X\,X^T\,\hat{A}^T$, applied identically to every colour column. This equals $\hat{A}^2$ only if $X\,X^T \approx I$, which needs an embedding dimension d at least as large as n. PI-GNN uses d far below n on most DIMACS graphs (for example 46 or 64), so $X\,X^T$ is a rank-d matrix and the $\hat{A}^2$ approximation must itself be checked. If the input embeddings are trainable there is an extra term, $(\hat{A}\,\hat{A}^T) \otimes (W^T W)$, acting across colours, so a node-space matrix alone is not the whole preconditioner. PI-GNN has two layers and a nonlinearity, so the linearisation is crude; 4B tests it, and rung D uses M_G as a fixed approximation.
- **What we do not assume.** We do not assume that M_G is low-pass or that it damps what colouring needs. $\hat{A}^2$ has gain λ^2, which is sign-blind: an eigenvalue of −1 is not damped. On random graphs it damps almost the whole spectrum except the top mode, and alternating modes are not singled out. What is supported is narrower: self-loops pull the alternating end of the spectrum toward zero (on an even cycle the gain of the most alternating mode falls from 1.0 to 0.11 with self-loops). These statements are about the $\hat{A}^2$ approximation; the actual induced preconditioner may differ.

- **What we measure instead.** (i) The spectrum of Â with and without self-loops, and of M_G (or of K on small graphs). (ii) The spectral content of centred one-hot colourings: project out the top eigenvector of Â, which is proportional to D^{1/2}·1 (subtracting the column mean is equivalent only on regular graphs), and do this for several good colourings (TabuCol, DSATUR, planted where available), reporting the spread, because different valid colourings load differently on the spectrum. (iii) The gain of M_G on the bands where those colourings carry their energy. The Jacobian Gram matrix is expensive, so compute K only on small graphs (n up to a few hundred). Optional extension if rung D shows an effect: a high-pass alternative such as $(I - Â)^2$.
- **4B, update alignment (gated).** At matched states, measure the cosine similarity between the actual logit update, the free-logit gradient direction and the preconditioned gradient direction. Run it with plain gradient descent and with the optimiser actually used (Adam or AdamW): Adam rescales updates per parameter, so a low alignment under Adam could reflect Adam rather than the preconditioning story. This tests the mechanism directly, without assuming a constant kernel.
- **4C, smoothness (always logged).** Track the Dirichlet energy of P over training for PI-GNN and free logits, together with Potts loss, hard conflicts and row entropy. Low Dirichlet energy together with poor colouring quality is evidence consistent with oversmoothing. High Dirichlet energy alone is not evidence of a good colouring, because unstructured random outputs also vary strongly across edges.
- **4D, targeted architecture checks (gated).** Self-loops on or off; orthogonal versus random input features. No depth sweep: Vanderbush & Weber already ran one, and it expands scope. The self-loop prediction is specific: under the $Â^2$ approximation, removing self-loops preserves the full gain of the alternating mode on a connected bipartite graph (eigenvalue −1), so it should help on bipartite or strongly alternating structure; we test whether this prediction survives for the actual induced preconditioner and for PI-GNN itself. Test it on a small designed family (even cycles, grids and bipartite random graphs with k = 2) and on a control family (dense random graphs near χ) where the prediction says self-loops matter little. Vanderbush & Weber's finding that self-loops hurt was on Erdős–Rényi graphs, where the alternating mode is not singled out; if self-loops hurt there too, the spectral explanation does not account for it, and we say so in advance.
- **4E, movement from initialisation (always logged).** Track parameter and logit distance from initialisation. The constant-kernel story needs lazy training; the instantaneous kernel of 4A is meaningful on any run but expensive. A narrow network with a large learning rate (Semyon's v3 configuration: hidden size 18, learning rate 0.0268) is likely to move far, in which case only the instantaneous version applies.

7. Secondary case studies

Case study A: explaining the warm-start no-op

Do not rerun a large warm-start grid. Use the existing results as the phenomenon to explain, and add only cheap diagnostics that connect to the core mechanism.

- **Phenomenon:** Semyon E3 (the TabuCol prior improved 0 of 20 graphs, tied 16, was worse on 4; over E2 and E3 together, 0 improvements and 7 regressions in 68 graph-tests). His prior of ε = 0.1 puts about 0.9 on TabuCol's colour, the regime where Vanderbush & Weber found no benefit over local search. HWS §3–§4 show the same copying behaviour for a supervised model.
- Check whether saved TabuCol colourings at k = χ−1 are already single-flip optimal. First check whether the colourings were saved; if not, a rerun on the 20 graphs is cheap.
- Measure the Hamming distance from the warm start during PI-GNN refinement, and whether movement correlates with improvement or regression.
- If a rerun is needed, use a small representative subset and one strong prior setting.

Interpretation: if the warm start is already a one-flip local minimum and PI-GNN barely moves, the earlier zero-improvement result becomes an explained consequence of the endpoint analysis rather than a disconnected negative experiment.

Case study B: lineage control

- Compare PI-GNN, free logits, Potts mean-field annealing, vertex descent with restarts and TabuCol on a representative subset at equal wall-clock budgets. Tune the annealing schedule on the tuning set, starting from the phase-transition prescription of Peterson & Söderberg.
- Purpose: not to claim mean-field annealing is new, but to test whether PI-GNN reaches qualitatively better minima than much simpler Potts-style optimisers.
- This is secondary and must not delay the three core experiments.

8. Supporting objective result

Potts versus Erdős is not a main branch. Existing controlled geocolor experiments indicate that the two losses tie (TC2: mean gap to χ 1.46 for Potts q-descent against 1.48 for Erdős; tempered Erdős with q-descent ties Potts, 5/79/6, p = 0.76), while q-descent and decoding have much larger effects. The Erdős colour term's gradient also shrinks like e^(−n/q), so Erdős reduces to the Potts conflict term when that term is effectively off. Use these results as motivation for studying parameterisation and decoding rather than proposing another loss.

The raw data of that comparison was lost. If it is needed for the final report, rerun a small confirmation with identical model, optimiser, budget, decoder, stopping rule and protocol.

9. Minimum viable project and execution order

The minimum viable project is exactly three core experiments: Experiment 1 (endpoint dynamics over training length), Experiment 2 (D-CE versus argmax plus vertex descent, with null and permuted-P controls) and Experiment 3 (PI-GNN versus tuned free logits, rungs A and B). Together they answer the

central question: does PI-GNN itself produce better Potts solutions, or do the gains come primarily from parameterisation and post-processing?

- **Stage 0, shared harness:** implement and save P at several budgets; standardise single-flip optimality, vertex descent, D-CE, timing and statistics.
- **Stage 1, cheap diagnostics:** reanalyse existing B5 outputs; run endpoint curves and decoder null controls.
- **Stage 2, central control:** implement and tune free logits, started from PI-GNN's exact initial logits; run A versus B (and B-rand) at $k = \chi - 1$, at matched steps and matched time, with the ungated logging.
- **Stage 3, gated mechanism:** only if A differs from B, add rungs C and D plus the gated diagnostics 4A, 4B and 4D.
- **Stage 4, supporting case studies:** the warm-start explanation and the mean-field comparison, if time remains.
- **Stage 5, confirmation:** repeat headline comparisons with 5 seeds on the final synthetic and DIMACS subsets and run the paired statistics.

10. Planned headline figures and tables

- **Figure 1:** training budget against the fraction of PI-GNN outputs that are single-flip optimal and the conflict reduction from vertex descent, with the joint $(\tilde{g}, \tilde{H})$ distribution as a companion panel.
- **Figure 2:** decoder comparison on the same frozen P, including the trained-P, permuted-P and null-P controls.
- **Figure 3:** PI-GNN versus free logits (same-start and random-start arms): conflicts and Potts loss against optimisation steps and against wall-clock time, with the Dirichlet-energy trace as a companion panel.
- **Conditional Figure 4:** update alignment (plain gradient descent and Adam), the spectrum of the induced preconditioner against the spectral profile of good colourings, or the PI-GNN, free-logit and fixed-preconditioner comparison, if the gated extension is triggered.
- **Table 1:** paired win/tie/loss and statistical tests for the three core comparisons.

11. Possible conclusions

- **A:** PI-GNN contributes little; free logits reach comparable minima and simple descent reproduces most of the decoder gain.
- **B:** PI-GNN matters, but a fixed graph preconditioner reproduces much of its effect.
- **C:** full PI-GNN consistently reaches better minima than both direct and fixed-preconditioned optimisation.
- **D:** decoding and local search dominate the differences between optimisation methods.

- **E:** the trained P is no more informative than a null or permuted P, so the decoder is effectively doing most of the solving.

Any of these is useful because the project is a mechanistic study, not a solver benchmark that only succeeds if PI-GNN beats TabuCol.

12. Where the codebases fit

- **geocolor:** the primary harness; the PI-GNN reimplementation and saved B5 outputs; the endpoint and decoder experiments; the likely home of the free-logit control.
- **Semyon's GraphExperiment:** existing PI-GNN, TabuCol and color6 results; warm-start case-study data; the optional mean-field and TabuCol lineage comparison.
- **HWS:** synthetic generators and supporting warm-start and inductive evidence. No new HWS model training is required.

13. Risks and stopping rules

- **Compute:** do not attempt 100,000 epochs on 48 graphs × 5 seeds × many methods. Use a small diagnostic subset to choose feasible tiered budgets, then report the budget curve honestly.
- **Initialisation confound:** free logits must start from PI-GNN's exact Z_0; otherwise a PI-GNN advantage cannot be attributed to the dynamics rather than the starting point.
- **Tuning fairness:** tune PI-GNN and free logits separately on the disjoint tuning set. A tuned free-logit baseline may beat PI-GNN with default hyperparameters; that is a legitimate result.
- **Scope creep:** Experiments 1–3 are sufficient for the course project. Preconditioning, warm starts and mean-field are extensions, not requirements.
- **Mechanism gate:** if PI-GNN and free logits are indistinguishable, do not spend time on elaborate kernel or preconditioner analysis. That null result is itself the central finding, and the ungated logging supplies its explanation.
- **Kernel cost:** the Jacobian Gram matrix K is expensive; restrict it to small graphs and use the cheaper update-alignment test (4B) elsewhere.
- **Spectral claims:** the self-loop prediction is only about bipartite or strongly alternating structure. If it fails on the designed family, the spectral explanation is wrong, and that is reported as a result.
- **Lost data:** the loss-comparison raw runs are gone, so any Erdős confirmation needs a rerun.
- **Novelty:** do not sell any single control as new. The contribution is the controlled decomposition and the diagnostic controls, especially null-P decoding, endpoint-versus-training-length, and the free-logit → fixed-preconditioner → PI-GNN ladder.

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Entries with an arXiv number were checked against arXiv. Confirm the venues of the others before submitting.